Posterior reversible encephalopathy syndrome in Guillain-Barré syndrome

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ABSTRACT

Autonomic dysfunction is a well-known complication of Guillain-Barré syndrome (GBS) and may manifest as hemodynamic fluctuations. Posterior reversible encephalopathy syndrome (PRES) is commonly associated with acute hypertension, but is rarely reported to occur in association with GBS. We describe a patient with GBS who developed PRES in the setting of autonomic dysfunction and review the clinical features of all 12 previously reported patients with co-occurrence of GBS and PRES. Almost all cases have occurred in women over the age of 55, raising the possibility of increased sensitivity to dysautonomia in this patient group.

1. Case report

A 63-year-old woman with a history of depression and anxiety presented with 3 days of back pain and cramps in her distal extremities. She had a self-limited gastrointestinal illness the week prior. On admission, her blood pressure and pulse were 172/98 mmHg and 96 beats per minute respectively. Shortly after presentation, she acutely developed bilateral loss of vision followed by a generalized tonic-clonic seizure. Lorazepam was administered and she was intubated for airway protection in the setting of post-ictal respiratory distress. MRI of the brain showed symmetric parieto-occipital T2-weighted hyperintense lesions, consistent with posterior reversible encephalopathy syndrome (PRES) (Fig. 1). Lumbar puncture demonstrated elevated total protein (59 mg/dl; normal range 5–55 mg/dl) and only one white blood cell. Levetiracetam was given and she was extubated within a day. Her electroencephalogram showed no epileptiform discharges. For intermittent hypertension, she was treated with labetalol. Over the next several hours, she developed progressive dyspnea and diffuse weakness. She was reintubated due to worsening respiratory failure. On examination, she was able to follow commands and mouth some words. Her extra-ocular movements were intact and her face was symmetric. Her muscle tone was diminished and no extremity movements were observed, either spontaneous or to noxious stimuli. Deep tendon reflexes were absent. Her blood pressure became more labile, ranging from 60/40 to 190/80 mmHg. Nerve conduction studies revealed a severe generalized polyneuropathy with absence of almost all compound muscle and sensory nerve action potentials. She was diagnosed with Guillain-Barré syndrome (GBS) and treated with intravenous immunoglobulin G (IVIG; 2 g/kg over 5 days). She had a prolonged hospitalization and required placement of gastrostomy and tracheostomy tubes. When seen in follow-up 10 months later, there was significant clinical improvement. She no longer required respiratory or nutritional support. She returned to all of her pre-hospitalization activities and was functionally independent. Her neurologic examination

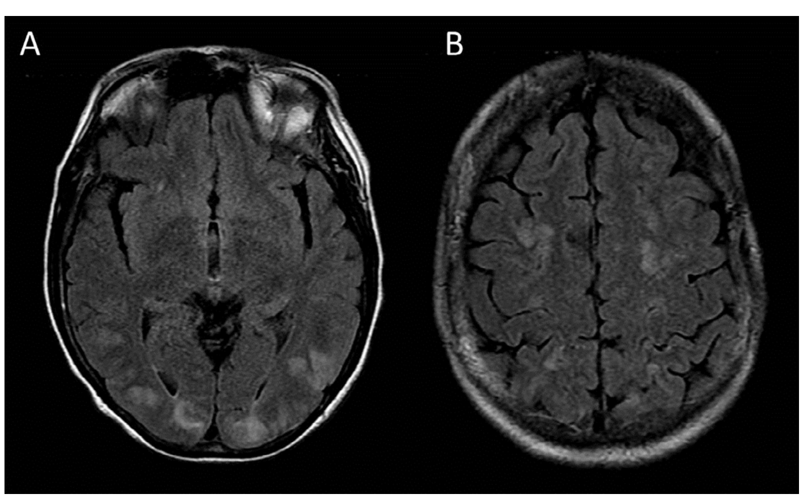

Fig. 1. Axial T2-weighted fluid attenuated inversion recovery MRI demonstrates bilateral symmetric hyperintense lesions involving mostly the parietal and occipital lobes (A) with some involvement of the frontal lobes (B).

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Table 1
Summary of reports of co-occurrence of posterior reversible encephalopathy syndrome and Guillain-Barré syndrome

Age	Sex	Initial GBS symptoms	Initial PRES symptoms	Days between GBS symptom onset and PRES symptom onset	History of HTN	Initial BP (mmHg)	CSF protein (mg/dl)	CSF WBC	EMG/NCS	MRI findings	Treatment	Outcome	Reference
58	F	Back/radicular pain, sensory changes in feet	Encephalopathy, seizure	3	No	152/82	93	2	CMAP amplitude ↓, velocity ↓, temporal dispersion, F wave absent. Blink reflex prolonged	PRES [^]	IVIG	7 months Walk independently	[4]
66	F	Back pain, ascending paralysis, R facial weakness	Encephalopathy	5	Yes	226/106 (maximum)	115	2	"Consistent with GBS"	PRES + basilar artery vasoconstriction, multiple infarcts, basal ganglia petechial and R frontal subarachnoid hemorrhages	IVIG	3 weeks Vision returned but quadriplegic	[5]
62	F	Back pain radiating to L shoulder	Seizure, encephalopathy	6	NR	191/98	112	"Normal"	CMAP latency ↑, SNAP L median absent, F wave prolonged	PRES	IVIG	1 month Full recovery (except reduced L ankle reflex)	[6]
62	F	Flaccid paralysis, facial diplegia	Cortical blindness	Simultaneous	NR	204/113	95	NR	CMAP latency ↑, conduction block. SNAP latency ↑, amplitude ↓, F wave prolonged	PRES + occipital and cerebellar infarcts, small hemorrhage	PLEX	9 months Full recovery (except absent ankle reflexes)	[7]
82	F	Back pain, leg weakness and paresthesias	Confusion, seizure	6	Yes	148/78	133	"Normal"	CMAP partial conduction block, temporal dispersion, F wave prolonged	PRES	PLEX	2 weeks Mental status normal, improved lower extremity strength	[8]
57	F	Sensory changes in fingers/feet	Headache	Simultaneous	No	180/100	147	5	"Demyelinating polyneuropathy"	PRES	IVIG	2 weeks Improved strength and sensation	[9]
67	F	Shoulder/back pain	Drowsiness, L homonymous hemianopia	3	No	190/90	131	0	CMAP latency ↑, amplitude ↓, SNAP latency ↑ or absent, F wave prolonged	PRES	IVIG	1 month Full recovery	[10]
28	F	Sensory changes in hands/feet	Asymptomatic	7 [*]	No	150/85	75	"Normal"	CMAP amplitude ↓, SNAP amplitude ↓, F wave prolonged	PRES	IVIG (stopped due to allergy)	5 months Moderate distal sensory loss, neuropathic pain	[11]
63	F	Sensory changes in feet	Headache, cortical blindness	4	No	180/100	189	0	CMAP latency ↑, velocity ↓, amplitude ↓, SNAP velocity ↓, amplitude ↓, F wave absent	PRES	IVIG, PLEX	1 year Walk with cane, R facial weakness, distal arm weakness	[12]
76	F	Radicular pain and paresthesias in all extremities	Encephalopathy, seizure	14	No	210/116	114	5	CMAP conduction block, SNAP amplitude ↓, velocity ↓, F wave absent	PRES	IVIG	1 month Full recovery (except diffuse areflexia, vibratory sensation loss in distal legs)	[13]
67	F	Neck pain, paresthesia	Asymptomatic	7 [*]	NR	NR	67	"Normal"	"Consistent with GBS"	PRES	IVIG	NR	[14]
63	M	Gait difficulty, arm weakness	Seizure	5	No	192/94	176	4	CMAP latency ↑, amplitude ↓, velocity ↑	PRES	IVIG	6 months Ambulatory, independent with ADL	[15]
63	F	Back pain, sensory changes in digits	Cortical blindness, seizure	3	No	172/98	59	1	CMAP absent, SNAP absent	PRES	IVIG	10 months Full recovery (except mild paresthesias, vibratory sensation loss in feet)	Present report

[*] MRI of the brain showing PRES was obtained 7 days after GBS symptom onset.
[^] MRI findings of PRES defined as relatively symmetric, posterior-dominant T2-weighted hyperintense lesions.
ADL = activities of daily living, BP = blood pressure, CMAP = compound muscle action potential, CSF = cerebrospinal fluid, EMG = electromyography, F = female, GBS = Guillain-Barré syndrome, HTN = hypertension, IVIG = intravenous immunoglobulin G, L = left, M = male, NCS = nerve conduction study, NR = not reported, PLEX = plasma exchange, PRES = posterior reversible encephalopathy syndrome, R = right, SNAP = sensory nerve action potential, WBC = white blood cell count, ↑ = increased, ↓ = decreased.

was normal with the exception of mild paresthesias in her fingertips and reduced vibratory sensation in her feet.

2. Discussion

Autonomic dysfunction is a well-described complication of GBS and affects over 60% of patients [1]. Manifestations include hypertension and tachycardia alternating with hypotension and bradycardia, arrhythmia, and urinary retention. Severe dysautonomia occurs in 20% of patients and can contribute to morbidity and mortality [2]. PRES is associated with a myriad of medical conditions and medications, but acute or relative hypertension is a common trigger [3]. PRES co-occurring with GBS is a rarely reported phenomenon, with 12 documented patients in the English literature to our knowledge (Table 1) [4–15].

Considering all reported cases, including our patient, 12 (92%) were female. Twelve (92%) were over the age of 55 with an average age of 63 (range 28 to 82 years). Nine (69%) had distinct onset of GBS-related symptoms (that is, back pain, sensorimotor deficits) prior to PRES-related symptoms (encephalopathy, visual changes, seizure). On average, GBS symptoms occurred 5 days (range 3 to 14) before PRES symptoms. Two patients (15%) were diagnosed with GBS and PRES simultaneously. Two patients (15%) had GBS but only incidental radiological evidence of PRES without clinical signs or symptoms of the disorder. Regarding GBS characteristics, all patients had cerebrospinal fluid albuminocytologic dissociation, with average total protein 116 mg/dl (range 59 to 189). Nerve conduction studies were performed in all patients and showed findings consistent with GBS. With respect to PRES characteristics, all patients were hypertensive at time of presentation, but only two (12%) had a history of chronic hypertension. MRI findings typical of PRES were seen in all patients. One patient also had occipital and cerebellar infarcts with a small hemorrhage while another had evidence of reversible vasoconstriction with associated infarcts and hemorrhage [5,7]. All patients were treated with IVIG and/or plasma exchange, but the majority (10, 77%) received IVIG alone. PRES was diagnosed prior to initiation of treatment in 12 patients (92%). Outcome was reported in 12 patients (92%) with an average follow-up time of 5 months (range 2 weeks to 1 year). Seven patients (54%) had recovery to functional independence. Of the five patients whose follow-up was longer than 6 months, four (80%) had near-complete recovery.

It has been proposed that GBS is an independent risk factor for PRES [6,7,12,15]. The exact mechanism for the relationship between the conditions is unknown, but one hypothesis is that dysautonomia may be a contributing factor. Blood pressure instability, in particular acute hypertension, may lead to increased capillary filtration pressure in the cerebral vasculature, resulting in secondary endothelial damage and development of PRES [3]. This is supported by the fact that all reported cases of GBS-associated PRES had early hypertension while only 17% had a history of chronic hypertension. Most patients did not develop symptoms of PRES until several days after onset of GBS symptoms, suggesting that duration as well as degree of hypertension may also play a role. An alternative theory suggests that the increased cerebrospinal fluid levels of inflammatory cytokines and chemokines seen in GBS may similarly alter capillary membrane permeability and disrupt the blood–brain barrier, resulting in PRES [2,16,17].

The majority of reported cases of GBS-associated PRES were women over age 55. Overall, men are about 1.5 times more likely than women to be affected by GBS [18]. Conversely, PRES is thought to occur more commonly in women, although robust epidemiological data are lacking [19]. This raises the possibility of sex and age-specific factors that may increase susceptibility to autonomic instability or its effects. The incidence of PRES co-occurring with GBS is unknown. Two reported patients had incidental discovery of radiographic evidence of PRES without clinical signs of the syndrome [11,14]. Therefore, subclinical impaired cerebrovascular autoregulation may be more common in GBS than has been documented, since most patients are unlikely to have an indication for brain MRI. Further prospective studies may be helpful in clarifying the true incidence of PRES occurring in the setting of GBS.

Conflicts of Interest/Disclosures

Dr. Chen, Dr. Kim, and Dr. Henderson report they have no financial or other conflicts of interest in relation to this research and its publication. Dr. Berkowitz reports no relevant disclosures, but receives royalties from *Clinical Pathophysiology Made Ridiculously Simple* (Medmaster, Inc.) and *The Improvising Mind* (Oxford University Press). This research received no grant funding.

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